



On using planning poker for estimating user stories

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ABSTRACT

While most studies in psychology and forecasting stress the possible hazards of group processes when predicting effort and schedule, agile software development methods recommend the use of a group estimation technique called planning poker for estimating the size of user stories and developing release and iteration plans. It is assumed that the group discussion through planning poker helps in identifying activities that individual estimators could overlook, thus providing more accurate estimates and reducing the over-optimism that is typical for expert judgment-based methods. In spite of the widespread use of agile methods, there is little empirical evidence regarding the accuracy of planning poker estimates. In order to fill this gap a study was conducted requiring 13 student teams to develop a Web-based student records information system. All teams were given the same set of user stories which had to be implemented in three Sprints. Each team estimated the stories using planning poker and the estimates provided by each team member during the first round were averaged to obtain the statistical combination for further comparison. In the same way the stories were estimated by a group of experts. The study revealed that students' estimates were over-optimistic and that planning poker additionally increased the over-optimism. On the other hand, the experts' estimates obtained through planning poker were much closer to actual effort spent and tended to be more accurate than the statistical combination of their individual estimates. The results indicate that the optimism bias caused by group discussion diminishes or even disappears as the expertise of the people involved in the group estimation process increases.

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1. Introduction

Poor estimation performance is an important concern in the software industry. A review of surveys on software effort estimation (Moløkken and Jørgensen, 2003) has shown that most projects (60–80%) encounter effort and/or schedule overruns, and that the estimation methods in most frequent use are expert judgment-based (Hughes, 1996). A further study dealing with expert estimation (Jørgensen, 2004) has revealed that there is no substantial evidence in favor of using estimation models, and that there are situations where we can expect expert estimates to be more accurate than formal estimation models. The study has also found that expert judgment-based estimates tend to be over-optimistic, especially when estimators estimate their own work. Therefore, it was recommended to continue with research on how to reduce the over-optimism bias and how to select and combine a set of expert estimates.

In order to provide a basis for the improvement of software estimation research a systematic review of software development

cost studies (Jørgensen and Shepperd, 2007) identified 304 journal papers describing research on software development effort or cost estimation, 61% of them dealing with the introduction and evaluation of estimation methods. The review revealed that expert judgment-based approaches are discussed in only 15% of the papers in spite of the fact that expert judgment-based approaches are, by far, the most commonly used approaches by the software industry. Additionally, the review identified a lack of in depth studies of the actual use of estimation methods and real-life evaluations. No study containing an in-depth data collection and analysis of how estimation methods were actually applied was discovered. Consequently, one of the clear findings was that the coverage of the research area should be increased by conducting more studies on estimation methods commonly used by the software industry.

A recent debate on software development effort estimation (Jørgensen and Boehm, 2009) revealed several reasons for focusing research on judgment-based methods, which (according to Jørgensen) the software industry is much more willing to accept and properly use than formal estimation models. Therefore, the software engineering discipline should become evidence-based and concentrate on improving those processes for which the evidence suggests that improvements will more likely occur.

Planning poker (Grenning, 2002) is one of the recent expert judgment-based estimation techniques that is widely used in agile

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methods for software development, especially Scrum (Schwaber and Beedle, 2002; Schwaber, 2004) and Extreme Programming (Beck and Andres, 2005). In both methods, a user story (Cohn, 2004) is the unit of which software features are estimated and developed. Each user story consists of three parts: a written description (used for planning and as a reminder), conversations about the story (to flesh out the details), and acceptance tests (to determine when a story is “done”). A story description is formulated in the language of the customer and is intentionally short enough to be hand-written on a paper note card. It serves merely as a reminder for conversations with the customer in order to clarify the story details and document the expectations of the project's users in the form of acceptance tests.

Strictly following Scrum, user stories are defined by the Product Owner (i.e., the customer representative responsible for maintaining and prioritizing all user stories that are currently known), while the estimates are made by the Scrum Team (i.e., the group of developers responsible for user stories implementation). Using planning poker, the Team allocates each story a number of story points representing the effort required for its implementation. A story point usually (although not necessarily) corresponds to an ideal day of work and often only a predefined set of possible values is used, e.g., 0.5, 1, 2, 3, 5, 8, 13, 20, 40 and 100 (Cohn, 2004, 2005).

Estimates are used during Release and Sprint planning meetings in order to create Release and Sprint plans, respectively. The Release plan is created by allocating stories to Sprints strictly considering their priority and the Team's estimated velocity, i.e., the number of story points that the team can implement during a Sprint. At the beginning of each Sprint the Team re-estimates its velocity and the remaining user stories in order to obtain a more realistic plan for subsequent iterations. During the Sprint, the team further decomposes each story into constituent tasks that must be performed to deliver required functionality by the end of the Sprint. The tasks emerge as the Sprint evolves and are divided so that each takes roughly 4–16 h to finish (Schwaber, 2004).

Planning poker is assumed to provide better estimates than individual experts by ensuring that all developers participate equally in the estimation process, regardless of whether they are among the loudest or most influential people in the Team. According to Moløkken-Østfold and Jørgensen (2004) participation of people with different backgrounds helps in reducing the over-optimism of expert judgment-based estimates by identifying more issues affecting implementation, and alleviating the problems of anchoring and strong personalities. On the other hand, evidence from other research communities (e.g., Armstrong, 2006; Brown, 2000; Buehler et al., 2005) points out the dangers of group processes, arguing that individuals are inherently optimistic and this optimism bias is increased by group interaction.

In order to contribute to the research of group estimation techniques we conducted a study on using planning poker for estimating user stories in a Scrum project requiring 13 student teams to develop a Web-based student records information system. The study was part of the capstone software engineering course that students take in their last (eighth) semester and lasts 15 weeks (Mahnic, 2012). All teams were given the same set of user stories that had to be implemented in three Sprints. At the beginning of the project each team had to estimate the stories using planning poker and prepare the Release plan. The statistical combination of individual estimates was computed simply as the average of the estimates provided by each team member in the first round of planning poker. During the course the amount of work spent was recorded in order to compare the actual effort to initial estimates. The same stories were also estimated by a group of experienced professionals, thus giving the possibility to compare estimation capabilities of beginners and experienced software developers.

The reminder of the paper is organized as follows: Section 2 describes related work and the relationship between planning poker and other group estimation techniques. In Sections 3 and 4, respectively, research questions and study design are described. The results are presented in Section 5 and discussed in Section 6. Section 7 provides a conclusion.

2. Planning poker and related group estimation techniques

Planning poker belongs to a set of group estimation techniques (Furulund, 2007; Moløkken-Østfold and Jørgensen, 2004) that build on the principle that more heads are better than one. User stories are estimated at an estimating session by members of the development team responsible for implementation. For each story, the Product Owner begins by explaining its requirements. In turn, the team discusses the work involved, posing questions to the Product Owner as needed. When the story has been fully discussed, each estimator privately estimates the required effort by writing the corresponding number of story points on a paper note card or choosing a card with the corresponding predefined value. All cards are revealed simultaneously in order to assure the independence between the different group members. If the estimates differ too much the estimators discuss their estimates. The high and low estimators should especially share their reasons. After discussion, the group revotes on their estimates by revealing again all cards at the same time. The process is repeated until consensus is achieved.

Planning poker is not completely new, but represents a variant of the Delphi technique (Brown, 1968; Dalkey, 1969; Linstone and Turoff, 1975), which was devised by the RAND corporation in the 1950s as a method of eliciting and refining group judgments. The Delphi technique has three features (Dalkey, 1969): (1) anonymous response, (2) iteration and controlled feedback, and (3) statistical group response. Estimates are obtained by anonymous expert interaction through several iterations, supervised by a moderator until an appropriate aggregate of individual opinions is obtained. Rowe and Wright (1999) provided a systematic review of empirical studies looking at the effectiveness of the Delphi technique and a critique of this research. Their findings suggest that Delphi groups outperform statistical groups and standard interacting groups, although the technique has shown no clear advantages over other structured procedures.

Boehm (1981) proposed a modification of the Delphi technique called Wideband Delphi. Wideband Delphi includes more group interaction, and reduces complexity and overhead of the original method. The experts provide their estimates anonymously, but meet to discuss (and possibly revise) them in the context of other estimates. The cycle repeats until the group decides that the average is representative and satisfactory. Wideband Delphi has been proposed as an estimation method in several books (e.g., Fenton and Pfleger, 1997; McConnell, 2006; Stellman and Greene, 2005) and professional articles (e.g., Wieggers, 2000). McConnell (2006) reports his experience with the use of Wideband Delphi suggesting that it cuts estimation error by an average of approximately 40% compared to the initial group average.

Moløkken-Østfold et al. (2008) classified group estimation techniques with regard to (1) their structure (the level of formality, amount of learning requirements, and degree of rigidity), (2) level of anonymity (whether the estimators are anonymous to each other), (3) amount of interaction among estimators, and (4) overhead (extra amount of effort spent on estimating). According to this classification the Delphi technique is the most complex, having heavy structure and major overhead. On the other hand, it assures full anonymity and no interaction among estimators, thus preventing potential undesired effects of group interaction. Wideband Delphi

reduces complexity of the Delphi technique and includes more group interaction. Its structure and overhead are moderate and offer limited anonymity and interaction. Planning poker is considered a light-weight technique with face-to-face interaction, limited overhead and no anonymity.

Furulund (2007) further discusses the aforementioned techniques from the viewpoint of preempting potential undesired effects of group interaction stating that all group processes encompass a trade-off between (1) exploiting advantages of combining estimates in groups, (2) preempting undesired group effects, and (3) minimizing the amount of overhead. Through anonymity one hopes that the different group members will provide their own opinion without being influenced by political, social or other considerations. The Delphi technique enables full anonymity, while Wideband Delphi and planning poker represent a compromise between protecting the independence of estimators and allowing face-to-face communication. Wideband Delphi does it through making the initial estimates anonymous, while in planning poker it is done through the simultaneous revealing of estimates.

Planning poker has been, to the best of our knowledge, the subject of only three published empirical studies. Haugen (2006) compared planning poker to unstructured group estimation and found that planning poker produced more accurate estimates than unstructured combination for familiar tasks, but increased estimation error when the team lacked experience with estimation of similar tasks.

The second and third studies (Moløkken-Østvold and Haugen, 2007; Moløkken-Østvold et al., 2008) concentrated on comparison between group consensus estimates obtained through planning poker and the statistical combination (the mean) of individual expert estimates. It was found that: (1) group consensus estimates were less optimistic than the statistical combination of individual estimates of the same tasks and (2) group consensus estimates were more accurate than the statistical combination of individual estimates of the same tasks. However, the authors themselves pointed out two possible limitations of their studies: (1) only one project team was studied and (2) planning poker was not used for estimating user stories, but for estimating tasks, which were much smaller than user stories and consequently easier to estimate. Additionally, the study started after the project team studied had been working on the project for several months; therefore, the tasks did not require implementation of requested features from scratch, but referred to changes of the system requested by the client. For all these reasons it was suggested that future studies might seek to complement the aforementioned findings by investigating projects with different constraints, and using planning poker for estimating user stories.

3. Research questions

There is no common agreement whether group estimation techniques reduce bias of expert judgment-based estimates toward optimism and provide more realistic estimates. Many studies on group decision making from other research communities suggest just the reverse: that an initially optimistic decision can be rendered even more optimistic by group discussion. For example, a set of studies by Buehler et al. (2005) requiring participants to predict when they would complete upcoming group projects found an optimism bias for both individual and group predictions. Predictions generated through group discussion were more optimistic than those generated individually since the group discussion heightened participants' tendency to focus primarily on factors promoting successful task completion, and this selective focus on "planning for success" enhanced their optimistic outlook.

Armstrong (2006) argues that face-to-face meetings harm forecasting and decision making, suggesting that other methods of aggregating individual opinions, such as using the average of a set of independent judgments or the Delphi technique provide more accurate estimates. According to him, there is not enough evidence for the claim that face-to-face interaction is beneficial in certain contexts, such as when it is important to gain commitment. Even some software management textbooks (e.g., Boehm, 1981; DeMarco, 1982; Fenton and Pfleeger, 1997) frequently warn about the possibility that group pressure could influence group decisions unreasonably, but omit valuable group work aspects such as motivation and information sharing when describing the separate estimating groups and group estimation techniques.

On the other hand, recent research on group processes in software estimation by Moløkken-Østvold and Jørgensen (2004) and Moløkken-Østvold et al. (2008) have shown that estimating software projects is a type of task that is suitable for group discussion. They stress the difference between ad hoc groups estimating tasks of which kind the participants might have little experience and groups of experienced professionals who are motivated to perform in a professional manner.

It can be argued that the planning poker estimation technique possesses several characteristics that to a great extent exploit the benefits of group work and face-to-face interaction: (1) estimates are determined by developers that will be responsible for implementation of the project, thus increasing their stake in the project and commitment to act professionally; (2) everybody's opinion is heard, thus avoiding the influence of dominant or most influential people; (3) all estimates are shown at the same time, thus avoiding the problem of anchoring; and (4) arguments for and against the highest and lowest estimate are discussed, thus providing opportunity to find as many issues that affect the implementation as possible. However, due to limited number of empirical studies further evidence is needed to confirm or refute hypotheses regarding its benefits in practical use.

In order to fill this gap we conducted a study similar to Moløkken-Østvold et al. (2008), but with some important differences: (1) the basic unit of estimation was a user story (not a task); (2) all user stories were estimated at the beginning of the project in order to define the Release plan (not in the middle of the project when developers have much more experience); and (3) the study involved 13 teams working on the same project, thus giving the possibility to compare estimates of different teams and obtain statistically relevant results.

For practical reasons the teams were composed of Computer Science students (since it is almost impossible to find several teams developing the same set of user stories in an industrial environment) who took the capstone course in software engineering in their last (eighth) semester, thus blurring the difference between students and young professionals.

The aim of the study was to answer the following research questions:

RQ1: How do planning poker estimates provided by student teams differ from the statistical combination of their individual estimates:

- RQ1.1: Are the planning poker estimates less optimistic than the statistical combination of individual estimates?
- RQ1.2: Are the planning poker estimates more accurate than the statistical combination of individual estimates?

RQ2: How do planning poker estimates provided by experienced professionals differ from the statistical combination of their individual estimates:

- RQ2.1: Are the planning poker estimates less optimistic than the statistical combination of individual estimates?
- RQ2.2: Are the planning poker estimates more accurate than the statistical combination of individual estimates?

RQ3: How do students' estimates differ from estimates made by experienced professionals?

- RQ3.1: Is the statistical combination of individual estimates provided by experienced professionals more accurate than the statistical combination of estimates provided by students?
- RQ3.2: Are the planning poker estimates provided by experienced professionals more accurate than the planning poker estimates provided by students?

The statistical combination of individual estimates, which the planning poker estimates were compared to, was obtained by simply averaging (for each team separately) the individual estimates obtained during the first round of planning poker considering the recommendation from Jørgensen (2005) that a simple average often works as the best method for combining estimates.

The first two research questions were similar to the research questions in Moløkken-Østfold et al. (2008), and our assumption was that we would be able to confirm their results, i.e., that planning poker reduces optimism and provides better results than the statistical combination of individual estimates. Therefore, with regard to research question RQ1 we expected the planning poker estimates provided by student teams to be less optimistic and more accurate than the statistical combination of their individual estimates. Similarly, with regard to research question RQ2 we expected the planning poker estimates provided by experienced professionals to be less optimistic and more accurate than the statistical combination of their individual estimates.

The aim of the third question was to explore differences between students and experienced professionals. It was expected that both kinds of estimates provided by experienced professionals (planning poker and statistical combination) would be less optimistic and more accurate than those of the students.

By answering these questions we wanted to obtain a more detailed insight into the planning poker process and find out whether the group discussion in the follow-up rounds during planning poker sessions contributes to better effort estimates.

4. Study design

4.1. Project setting

The study required 13 student teams to develop a Web-based student records information system covering enrollment, examination applications, examination records, some statistical surveys, and a special module for the maintenance of all data required for the proper functioning of the system (i.e., the maintenance of various code tables, lists of required and optional courses, data about teachers of each course, etc.). Each team consisted of four students and acted as a self-organizing and self-managing Scrum Team collectively responsible for the implementation of the required functionality.

The project lasted 15 weeks and was divided into four Sprints. The first Sprint, Sprint 0, was a preparatory Sprint during which students received formal training on Scrum, user stories, and agile estimation and planning techniques (Step A; Table 1). At the end of Sprint 0 all teams were presented with the same initial Product Backlog consisting of 60 user stories (Step B). 55 stories

Table 1
Study overview.

A	Sprint 0	Formal training on Scrum and user stories
B		Initial Product Backlog presented
C		User stories estimated using planning poker
C.1		User story requirements presented by the Product Owner
C.2		The work involved discussed by each team
C.3		Individual estimates from the first round recorded
C.4		Group consensus estimate recorded
D		Expert estimates obtained using planning poker
D.1		User story requirements presented by the Product Owner
D.2		The work involved discussed by the team of experts
D.3		Individual expert estimates from the first round recorded
D.4		Group consensus expert estimate recorded
E		Velocity of each team estimated
F		Release plan of each team defined
G	Sprints 1, 2, and 3	Sprint planning meeting
G.1		New (and remaining) stories estimated
G.2		Sprint Backlog defined
G.3		User stories decomposed into constituent tasks
G.4		Tasks assigned to team members
G.5		Tasks estimated
H		Daily Scrum meetings
H.1		Effort spent on each task recorded
H.2		Amount of work remaining on each task recorded
I		Sprint review meeting
I.1		User story implementation evaluated by the Product Owner
I.2		A new story defined for each uncompleted/rejected user story
J	End of study	Actual effort computed on the basis of effort spent recorded at the Daily Scrum meetings

described the required functionality for four different user roles (student records administrative staff, students, teachers, and data administrator), whereas five stories described constraints that had to be obeyed (e.g., the system had to enable remote access to data through the Internet, all outputs should also be printable, and so on).

The initial Product Backlog was prepared by a teacher who had considerable experience in developing university information systems, thus being able to provide realistic user requirements and play the role of Product Owner. Each story contained a short description and a set of acceptance tests that had to be used to demonstrate that the story had been correctly and fully coded. The stories were divided into four groups on the basis of priority. Among the 55 stories describing the required functionality there were 21 “must have”, 4 “should have”, 3 “could have”, and 27 “won’t have this time” stories. The teams were asked to implement at least all “must have” and as many as possible “should have” stories for the first release. If they planned to be more productive they could also include the “could have” and even some of the “won’t have this time” stories. However, the “won’t have this time” stories were specified merely to illustrate the desired functionality for the next release.

In order to prepare the Release plan each team was required to estimate the “must have”, “should have”, and “could have” stories in story points, using the planning poker technique. The estimates were constrained to specific predefined values of 0.5, 1, 2, 3, 5, 8, 13, and 20, representing the number of ideal working days needed to fully implement each story. Since there were no epics (i.e., large stories that should have been split into smaller ones) only estimates in the range between 0.5 and 8 were expected. Nevertheless, the values of 13 and 20 were also allowed if a team considered a story more complex and difficult to implement.

All participants in the study were especially instructed to consider an ideal working day to be equal to 6 h of effective work in order to be able to convert hours actually worked back to story points and compare the estimates with actual effort spent. The conversion factor of 6 was chosen for two reasons: (1) because within the Agile for Scrum project management tool, which all teams used for maintaining project data, an ideal working day is by default 6 h and (2) because the company studied in [Moløkken-Østvold et al. \(2008\)](#) also deemed a full day of work to be 6 h.

Estimation (Step C) took place at a special meeting at which, in turn, the Product Owner presented the requirements of each story (Step C.1), and the teams discussed the work involved in fully implementing and testing the story until they believed that they have enough information to estimate the effort (Step C.2). Each team member then privately and independently estimated the effort. All estimates were revealed simultaneously and recorded to compute the statistical combination (the mean) of individual estimates for further comparison with the planning poker estimates (Step C.3). In case of disagreement, the team members with the lowest and highest estimate were required to explain their estimates to the team and a discussion ensued until the group was ready to revote on their estimates. If necessary, more estimation rounds took place until the teams came to a consensus on the estimate (Step C.4).

In order to compare students' estimates to expert estimates the same stories were also estimated by a group of four experts consisting of instructors and developers that had been involved in development of the University of Ljubljana student records information system (Step D). Again, the individual estimates of the first round (Step D.3) and the final planning poker estimate of each story (Step D.4) were recorded for the needs of the study.

The rest of the study consisted of three regular Scrum Sprints. It was assumed that the first release would be completed by the end of Sprint 3. Before Sprint 1 each team had to estimate its velocity (Step E) and develop its own Release plan (Step F) by allocating the stories into three Sprints so that the sum of story points fit within the capacity determined by the velocity estimate. At the beginning of each Sprint (Step G) the teams had to define the Sprint Backlog (Step G.2) by further decomposing user stories into constituent tasks and assigning responsibility for each task (Steps G.3 and G.4). Each team member individually estimated how many hours it would take to accomplish each task he/she had accepted (Step G.5).

For the purpose of the study, the teams had to meet regularly at Daily Scrum meetings (Step H) and collect data about the effort spent. At the Daily Scrum meeting each team member had to record the number of hours spent (Step H.1) and the amount of work remaining for each task he/she was responsible for (Step H.2). Since it was impossible to expect students to work on the project every day, two Daily Scrum meetings per week were prescribed, one on Monday and the other on Thursday.

When a team finished a story (or at latest at the Sprint review meeting), the Product Owner was asked to evaluate its implementation (Step I.1). The Product Owner strictly enforced the concept of "done", rejecting all stories that did not conform to user requirements. If a story was not "done" by the end of the Sprint a new story was defined in the Product Backlog requiring the completion of the missing features in one of the remaining Sprints (Step I.2). The newly defined stories were treated in the same way as the original stories from the initial Product Backlog. They were estimated using planning poker and considered for implementation in the next Sprint (Step G.1). Effort spent on each such story was recorded at the Daily Scrum meetings and added to effort spent on the original story from which it was derived.

Data relevant to the study were collected in Steps C.3, C.4, D.3, D.4, and H.1. At the end of the course the total number of hours spent on all tasks belonging to each story was converted into story

points (Step J), considering an ideal day of work to be 6 h as agreed before estimation took place in Step C.

4.2. Calculation of estimation accuracy

To calculate estimation accuracy the balanced measure of relative error (BRE) was employed, which is calculated as:

$$\text{BRE} = \frac{|\text{actual effort} - \text{estimated effort}|}{\min(\text{actual effort}, \text{estimated effort})}$$

The decision to use BRE as our key measure arose from the fact that during the last decade the measure traditionally used to measure the accuracy of effort estimations in software engineering, the magnitude of relative error of the estimate ($\text{MRE} = |\text{actual effort} - \text{estimated effort}| / \text{actual effort}$), came under sharp criticism by several researchers ([Foss et al., 2003](#); [Jørgensen, 2007](#); [Kitchenham et al., 2001](#); [Miyazaki et al., 1991](#); [Shepperd et al., 2000](#)). The main concern in our case was the fact that the use of MRE would cause uneven treatment of overestimated and underestimated stories, with underestimation being weighted insufficiently. BRE is a more balanced measure, evenly balancing overestimation and underestimation. Another reason for using BRE was compatibility with previous related research. Many recent studies on software effort estimation use BRE as the key measure (e.g., [Moløkken-Østvold and Jørgensen, 2005](#); [Moløkken-Østvold and Furulund, 2007](#)), including related studies of planning poker by [Moløkken-Østvold and Haugen \(2007\)](#) and [Moløkken-Østvold et al. \(2008\)](#). The decision to use BRE made it possible to directly compare our results to the results of previous research of the planning poker technique.

In order to measure the bias toward optimism or pessimism, the BREbias was calculated, which measures both the size and the direction of the estimation error:

$$\text{BREbias} = \frac{(\text{actual effort} - \text{estimated effort})}{\min(\text{actual effort}, \text{estimated effort})}$$

Because our BRE and BREbias values exhibited deviations from normality in terms of skewness (larger than ± 1) and kurtosis (larger than $+2$) we did not analyze our data with parametric *t*-tests as was done in the comparable research by [Moløkken-Østvold et al. \(2008\)](#), but used the most appropriate non-parametric alternative on the non-transformed data ([Darlington, 2011](#); [Lowry, 2011](#)). Research questions RQ1 and RQ2 required the before-after analysis of a single sample, while research question RQ3 required the analysis of two independent samples. Therefore, the non-parametric Wilcoxon signed-rank test ([Wilcoxon, 1945](#)) was used to answer RQ1 and RQ2, and the Mann-Whitney-Wilcoxon test ([Mann and Whitney, 1947](#)) to answer RQ3. A two tailed *p*-value of 0.05 or less had to be reached in order to demonstrate that the distributions of the two tested variables differ significantly from each other.

Since the expert group did not develop the system, the BRE and BREbias values of expert group were computed for each story separately by comparing the experts' estimate (of that story) to the arithmetic mean of the actual effort spent by the student teams (on that story). We found this decision reasonable because the experts were not able to provide their estimates for each team separately, but could only estimate the most likely effort of an average student team since they were unfamiliar with the characteristics of each particular team at the time of estimation. The actual effort spent by an average student team is therefore best represented by averaging the effort spent by all teams participating in the study for each story separately.

In contrast to [Moløkken-Østvold et al. \(2008\)](#), we did not use Cohen's size of effect measure (*d*) ([Cohen, 1988](#)) to express the effect size of the difference between tested variables since

it is sensitive to the presumption of normality which in our case was violated. Instead, we chose the non-parametric measure of the effect size Cliff's delta (δ) (Cliff, 1996), which is considered a robust and intuitive alternative to Cohen's d in situations where data are either non-normal, or are ordinal and therefore have reduced variance (Hess and Kromrey, 2004; Ledesma et al., 2009).

5. Results

In total, 28 user stories describing functionalities of different complexity were estimated at the beginning of the study by 13 student teams and a group of four experts. The teams were asked to implement at least 21 “must have” stories, and as many as possible of the 4 “should have” stories for the first release. However, some teams did not fulfill their release plans completely by the end of the course. Two teams were behind the plan because some of the part-time students were not able to work on the project consistently due to their job commitments. Additionally, a couple of stories developed in the last Sprint by some other teams were rejected by the Product Owner for not being completely “done”, resulting in another team that implemented less than 20 stories.

In order to obtain a homogeneous sample for statistical analysis consisting of the same 20 stories being successfully implemented by all teams we excluded the three aforementioned teams and proceeded with the analysis of 20 stories that were successfully developed by all of the remaining 10 teams.

Basic descriptive statistics for the final student teams sample and the experts sample are presented in Table 2.

5.1. RQ1: How do planning poker estimates provided by student teams differ from the statistical combination of their individual estimates?

Table 3 presents results of Wilcoxon signed-rank tests that we performed in order to answer the research question RQ1. For each test all data reported by SPSS are shown as well as the effect-size measure Cliff's delta.

In order to answer RQ1.1 we first tested both methods for bias toward optimism or pessimism and then analyzed the difference between the BREbias statistics.

Bias toward optimism/pessimism was analyzed by comparing the BREbias of the statistical combination of individual estimates and the BREbias of the planning poker estimates to a hypothetical unbiased BREbias distribution using Wilcoxon signed-rank test. The results are presented in the first two rows of Table 3. The BREbias medians of both kinds of estimates pointed to over-optimism (median BREbias of the statistical combination of individual estimates = 0.0519, median BREbias of the planning poker estimates = 0.0833); however, a statistically significant bias toward optimism was only confirmed for the planning poker estimates ($p = 0.034$). Even though the planning poker distribution differed statistically significantly from the hypothetical unbiased distribution, Cliff's delta of the tested difference was very small (-0.095).

Wilcoxon signed-rank test was also used to compare the BREbias of both kinds of estimates. Results are presented in the third row of Table 3 and indicate that planning poker did not reduce the optimism of initial individual estimates. On the contrary, if we turned the research question around and asked ourselves whether the planning poker estimates were more optimistic than the statistical combination of individual estimates, the answer would be affirmative since the single-sided Wilcoxon signed-rank test p -value

would be 0.0315. Cliff's delta of the tested difference is however very small (0.039).

In order to answer RQ1.2 another Wilcoxon signed-rank test was performed that compared the BRE of planning poker estimates to the BRE of the statistical combination of students' individual estimates (see the fourth row in Table 3). The results revealed statistically significant differences between the distributions of the planning poker estimates (median BRE = 0.6667) and the statistical combination of individual estimates (median BRE = 0.600). Thus, based on the p -value of 0.008 we can conclude that the planning poker estimates were less accurate than the statistical combination. However, the size of effect was again very small (0.0329).

5.2. RQ2: How do planning poker estimates provided by experienced professionals differ from the statistical combination of their individual estimates?

The results pertaining to research question RQ2 were obtained in the same way. First, the planning poker estimates and the statistical combination of individual experts' estimates were tested for bias toward optimism or pessimism; then, the BREbias and BRE statistics of both kinds of estimates were compared. The results of the corresponding Wilcoxon signed-rank tests are presented in Table 4.

It is interesting that the experts over-estimated the effort that was actually needed for the implementation. Statistical tests showed that the over-estimation was not statistically significant (p -values 0.086 and 0.173), but quite substantial (Cliff's delta 0.50 and 0.20). There are several possible explanations of this unexpected phenomenon. It seems that the experts – not estimating their own effort, but trying to adapt the estimates to students' abilities – increased the estimates too much. However, it can also be argued that the experts' estimates were quite realistic, since solutions provided by some teams – although formally offering the required functionality – lacked professional polish and would need additional effort to bring the code up to a useful, real-world level, which could survive an encounter with end users. Another, although less plausible explanation is that some teams reported less effort than was actually spent.

The planning poker estimates (median BREbias = -0.2231) were less pessimistic than the statistical combination of individual estimates (median BREbias = -0.2445) and more accurate (median BRE = 0.3260 vs. median BRE = 0.3835). However, statistical tests found these differences neither statistically significant nor having a substantial effect-size.

5.3. RQ3: How do students' estimates differ from estimates made by experienced professionals?

In order to analyze how students' estimates differ from estimates made by experienced professionals two Mann–Whitney–Wilcoxon tests were performed. The first one compared estimates obtained by statistical combination, while the second one referred to estimates obtained through planning poker. Results are presented in Table 5.

Both tests have shown a statistically significant difference between students and experts for both kinds of estimates (p -values 0.041 and 0.013). The effect-size of the difference can be considered medium (Cliff's delta 0.277 and 0.335). These findings – together with results of research questions RQ1 and RQ2 – indicate that there is an important difference in the way that planning poker affects estimates of experienced professionals in comparison to inexperienced beginners.

Table 2
Statistics.

	BRE students statistical combination	BRE students planning poker	BREbias students statistical combination	BREbias students planning poker	BRE experts statistical combination	BRE experts planning poker	BREbias experts statistical combination	BREbias experts planning poker	Test value = 0
<i>N</i>									
User stories (students)	200	200	200	200	200	200	200	200	200
User stories (experts)	20	20	20	20	20	20	20	20	20
Mean	1.0225	1.1074	.0778	.1802	.4210	.4286	–.1511	–.1424	.0000
Median	.6000	.6667	.0519	.0833	.3835	.3260	–.2445	–.2231	.0000
Std. deviation	1.29359	1.38715	1.64867	1.76752	.33084	.39649	.52153	.57381	.00000
Skewness	3.459	3.026	–1.138	–1.096	1.630	1.694	1.482	.532	
Std. error of skewness	.172	.172	.172	.172	.512	.512	.512	.512	.172
Kurtosis	19.183	14.020	11.432	9.215	3.742	2.666	3.492	2.722	
Std. error of kurtosis	.342	.342	.342	.342	.992	.992	.992	.992	.342
Range	11.00	11.00	17.52	17.00	1.41	1.40	2.38	2.86	.00

Table 3
RQ1 results.

	<i>N</i>	Mean rank	Sum of ranks	<i>Z</i>	<i>P</i> -value (2-tailed)	Cliff's delta
Test value = 0 – BREbias statistical combination						
Negative ranks	104	96.85	10,072			
Positive ranks	87	94.99	8264			
Ties	9			–1.182	0.237	–0.085
Total	200					
Test value = 0 – BREbias planning poker						
Negative ranks	102	99.48	10,146.5			
Positive ranks	83	85.04	7058.5			
Ties	15			–2.117	0.034	–0.095
Total	200					
BREbias planning poker – BREbias statistical combination						
Negative ranks	68	62.14	4225.50			
Positive ranks	75	80.94	6070.50			
Ties	57			–1.859	0.063	0.039
Total	200					
BRE planning poker – BRE statistical combination						
Negative ranks	58	66.16	3837			
Positive ranks	85	75.99	6459			
Ties	57			–2.642	0.008	0.0329
Total	200					

Table 4
RQ2 results.

	<i>N</i>	Mean rank	Sum of ranks	<i>Z</i>	<i>P</i> -value (2-tailed)	Cliff's delta
Test value = 0 – BREbias statistical combination						
Negative ranks	5	29,526	59			
Positive ranks	15	40,734	151			
Ties	0			–1.717	0.086	0.50
Total	20					
Test value = 0 – BREbias planning poker						
Negative ranks	8	20,668	68.5			
Positive ranks	12	29,160	141.5			
Ties	0			–1.363	0.173	0.20
Total	20					
BREbias planning poker – BREbias statistical combination						
Negative ranks	5	5.00	25.00			
Positive ranks	5	6.00	30.00			
Ties	10			–0.255	0.799	0.0575
Total	20					
BRE planning poker – BRE statistical combination						
Negative ranks	5	29,342	29			
Positive ranks	5	43,952	26			
Ties	10			–0.153	0.878	–0.052
Total	20					

Table 5
RQ3 results.

	N	Mean rank	Sum of ranks	Mann–Whitney U	Wilcoxon W	Z	P-value (2-tailed)	Cliff's delta
BRE students statistical combination – BRE experts statistical combination								
Students	200	113.28	22,655	1,445,000	1,655,000	–2.045	0.041	0.277
Experts	20	82.75	1655					
Total	220							
BRE students planning poker – BRE experts planning poker								
Students	200	113.86	22,772.5	1,327,500	1,537,500	–2.480	0.013	0.335
Experts	20	76.88	1537.5					
Total	220							

6. Discussion

6.1. RQ1: How do planning poker estimates provided by student teams differ from the statistical combination of their individual estimates?

With regard to RQ1, statistical tests presented in Table 3 indicate that (1) the statistical combination of individual estimates provided by student teams is less optimistic than the planning poker estimates and (2) that the statistical combination of individual estimates provided by student teams is more accurate than the planning poker estimates. However, the size of effect is very small.

From this point of view, the results concerning student teams are in line with findings reported by Buehler et al. (2005) that both individual and group predictions have an optimism bias which is further accentuated by group discussion.

The results are also in line with Jørgensen's claim (Jørgensen, 2005) that taking a simple average often works as the best method for combining estimates. Although the estimates obtained by computing a simple average of individual students' estimates also tended to be optimistic, they appeared to be more accurate than the estimates that the student teams provided using planning poker.

On the other hand, the results of statistical tests in Table 3 are in contrast with the findings of Moløkken-Østfold et al. (2008) that group discussion through planning poker can reduce optimism in software development projects. It seems that in spite of the fact that a typical software estimation task differs from tasks studied in other research areas, factors that promote the successful completion of tasks identified by Buehler et al. (2005) prevail when less experienced software developers are used as estimators.

Another possible reason for less accurate planning poker estimates provided by student teams lies in the fact that the teams consisted of members with the same background and knowledge, thus having a limited ability of exploiting group discussion as a means of identifying forgotten tasks that were not taken into account in initial estimates.

A possible explanation of less accurate planning poker estimates might also be that Computer Science students treat administrative tasks like estimating and planning as less important than other more technical activities. Consequently, they possibly did not put enough effort in group discussion. In a survey about the importance of twelve typical Scrum practices for a successful software development project that we conducted among students at the end of the study, the accuracy of planning poker estimates was ranked eleventh, i.e., next to last (Mahnic, 2011).

6.2. RQ2: How do planning poker estimates provided by experienced professionals differ from the statistical combination of their individual estimates?

The planning poker estimates provided by a group of experts tended to be less pessimistic and more accurate than the statistical combination of individual estimates. However, the size of effect

was very small and statistical tests found no significant difference between both kinds of estimates.

Although we agree with Moløkken-Østfold et al. (2008) that group discussion through planning poker can identify forgotten tasks and increase the initial estimates, the study has shown that the lowest and the highest estimates provided during the first round of planning poker act as strong anchors representing the lower and the upper boundary of the interval within which the group consensus is sought. We found no story having the final planning poker estimate outside these boundaries. These findings lead to the hypothesis that in most cases we cannot expect a significant difference between the planning poker estimates and the statistical combination of individual estimates. However, this hypothesis should be confirmed or refuted through a larger set of studies.

On the other hand, results of statistical tests also disagree with Armstrong's claim that face-to-face meetings are harmful for decision making (Armstrong, 2006). Considering the fact that planning poker estimates tended to be slightly better than the statistical combination, it seems reasonable to continue the research on group processes in software estimation started by Moløkken-Østfold and Jørgensen (2004) and Moløkken-Østfold et al. (2008) and seek special contexts in which the advantages of planning poker prevail.

In order to compare the planning poker estimates to the statistical combination of individual estimates more exactly we suggest not constraining planning poker estimates to specific predefined values, but allowing an arbitrary integer value representing hours instead of ideal days to be assigned to each story. Constraining planning poker estimates to specific predefined values can increase the estimation error and skew the comparison between both approaches.

6.3. RQ3: How do students' estimates differ from estimates made by experienced professionals?

Statistical tests revealed a statistically significant difference of medium effect-size between estimates provided by students and estimates provided by experienced professionals. Both kinds of estimates provided by the group of experts were statistically significantly more accurate than their counterparts provided by students.

Comparison between students and experienced professionals helped us to explain the mixed results of previous studies analyzing the effects of group discussion on estimation accuracy. Results of our study indicate that the optimism bias caused by group discussion diminishes or disappears by increasing the expertise of people involved in the group estimation process. It seems that inexperienced software developers behave similarly to ad hoc groups performing estimates of tasks they are not familiar with (like in studies conducted by Buehler et al., 2005), while highly motivated professionals with different project experiences and knowledge are able to take advantages of group discussion to analyze the problem from different perspectives, thus increasing both the possibility to identify forgotten tasks, and awareness that activities may be more complicated than initially thought (like in study conducted by Moløkken-Østfold et al., 2008).

6.4. Study validity

The conclusions of our study are based on the results of statistical tests which (1) exposed statistically significant differences between different treatments and (2) measured the size of effect. Non-parametric tests were used since the data did not support the presumption of a normal distribution. With regard to students the results have shown statistically significant over-optimism of their planning poker estimates indicating a shift toward increased optimism after the group discussion. However, the magnitude of the shift measured by the effect size was very small. With regard to experts the statistical analysis has confirmed that their estimates (planning poker and statistical combination) are significantly more accurate than those provided by students. The size of effect of the difference was medium (planning poker Cliff's delta 0.335, statistical combination Cliff's delta 0.277).

In order to assess the impact of possible confounding factors, Grimstad and Jørgensen's framework for analyzing the accuracy of software estimation (Grimstad and Jørgensen, 2006) was used, which provides a checklist for identifying factors with a potential impact on the measured estimation error. The factors are divided into three categories: (1) estimation ability factors (expert judgment skills, accuracy of an estimation model, skills in selection of estimation method, skills in the use of a formal estimation model), (2) estimation complexity factors (project management cost control ability, project member skill, client and sub-contractor performance, completeness and certainty of information, inherent project execution complexity, project priorities, flexibility in product and process execution), and (3) measurement process factors (inconsistent use of terminology, logging problems, difference between planned and actual output/process, measurement selection bias).

Considering research questions RQ1 and RQ2 in isolation we can conclude that none of the factors in the checklist, except the estimation method used (i.e., the factor we wanted to study), affected the results of our study. On the other hand, with regard to research question RQ3 the checklist indicates that besides the expert judgment skills (i.e., the factor we wanted to study), the project member skills and logging problems are the factors that could have affected the difference between students' and experts' estimates.

With regard to the project member skills, Grimstad and Jørgensen's checklist says that it is usually easier to estimate the effort of skilled developers, while the effort of unskilled developers is less predictable. In our study, this factor possibly decreased the accuracy of estimates provided by the group of experts when estimating the effort of student teams, which resulted in over-pessimistic estimates. However, it did not affect the conclusions derived from the results of statistical tests. The tests have shown that the experts' estimates were still significantly more accurate with medium effect-size than the estimates provided by student teams (in spite of the possibly decreased accuracy). It can be argued that the experts' estimates would have been even more accurate had they estimated the effort of skilled project members.

With regard to logging problems one might argue that some student teams did not record their effort conscientiously enough, thus representing less effort than actually spent. However, this problem also did not affect the final conclusions. On the contrary, the assumption that the amount of effort actually spent was greater than recorded additionally emphasizes the over-optimism of students' estimates and reduces the error of experts' estimates.

In order to increase construct validity, ten teams were studied working on the same problem. All teams estimated the same set of user stories; therefore, their estimates are directly comparable. A possible inconsistency of estimates in case of multiple estimations (Grimstad and Jørgensen, 2007) was eliminated by assuring that each story was estimated only once using planning poker, while the

statistical combination was computed automatically as a mean of individual estimates given in the first round. Additionally, in contrast to Moløkken-Østvold et al. (2008) planning poker was used for estimating user stories for which it is most commonly recommended (Cohn, 2005). The size of user stories was appropriate for estimation, ranging from 0.5 to 8 story points.

In spite of the fact that the study was conducted within the framework of a capstone software project, every effort was made to increase its external validity by simulating an industrial environment as closely as possible. User stories were defined on the basis of the student records information system that is actually used at the University of Ljubljana and the students were required to produce fully tested code that could survive the encounter with end users. Nevertheless, the main threat to external validity remains that only one project was used and only one group of experts studied. The experts did not estimate their own work, but the most likely effort of student teams. On the basis of statistical analysis, the results can be generalized only to students of the last year of Computer Science working on similar projects that require the development of administrative information systems, while more studies are needed on real projects of different size and complexity in order to generalize the findings to industry.

Since the planning poker estimation technique – as proposed by Grenning (2002) and Cohn (2005) – requires the development team to estimate their own effort, additional factors regarding the team composition should be considered. It is important to take into account not only motivation (Brown, 2000) and diversity of knowledge (Surowiecki, 2004), but also how the mix of experienced and inexperienced team members working on the same project affects the accuracy of estimates. Does the over-optimism of inexperienced developers prevail or does the knowledge and ability of experienced ones to discuss all the implications, identify forgotten tasks, and expose different alternatives lead to more realistic estimates? It would also be interesting to further investigate how the lowest and highest estimates provided during the first round of planning poker affect the interval within which group consensus is sought.

Our study provides a reasonable explanation of mixed results of previous research on group estimation processes, indicating that planning poker can be useful when used by experienced professionals who are able to take advantages of the follow-up rounds within planning poker sessions for discussing issues that affect the estimate of the most likely effort. Considering the aforementioned limitations, the results of our study can be used together with other studies as a stepping stone to further research, narrowing the focus to more experienced expert groups and searching for contexts where planning poker improves estimating accuracy by increasing commitment, sharing estimating expertise, promoting team growth, and refining solution understanding.

7. Conclusions

In spite of the fact that planning poker is widely used for estimating user stories in agile software development methods, there are almost no empirical studies investigating this technique in more detail. We wanted to fill this gap by conducting a study similar to that of Moløkken-Østvold et al. (2008), but involving several teams working on the same project and using user stories (not tasks) as the basic estimation unit. Additionally, the study design allowed for the possibility of comparing the estimates provided by students to estimates provided by experienced professionals, thus making it possible to clarify contradictory results of previous studies dealing with the impact of group discussion on the accuracy of effort estimates. While Buehler et al. (2005) found that group discussion made individual optimistic estimates even more

optimistic, Moløkken-Østfold et al. (2008) tried to prove that the use of planning poker in estimating software development effort reduces over-optimism and provides more accurate estimates than the statistical combination of individual experts' estimates.

The results of our study clarify findings of both aforementioned studies by showing that the optimism bias caused by group discussion diminishes or disappears by increasing the expertise of people involved in the group estimation process. Student teams exhibited similar behavior as the ad hoc groups studied by Buehler et al. (2005), providing planning poker estimates that were statistically significantly more optimistic than the statistical combination of their individual estimates. On the other hand, the analysis of experts' estimates yielded similar results as Moløkken-Østfold et al. (2008). The bias toward over-optimism disappeared and the planning poker estimates tended to be more accurate than the statistical combination. Although it was not statistically confirmed that planning poker estimates are more accurate, it seems reasonable to continue research on group processes in software estimation in order to find contexts in which the advantages of planning poker prevail over the statistical combination of individual estimates.

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